Poly-substance abuse, impulse control disorders, Forensic psychiatry

&

Vegetative Disorders

Substance abuse disorders

Clinical definitions

- **Abuse:** Use of drug in a manner deviant from society.
- **Dependence:** repeated use of drug with altered physiologic state, cessation results in specific syndrome.
  - **Psychological dependence:** cessation cause **craving** (severe urge or compulsion to obtain drug) even when aware of damage it cause and **decreased functioning**
  - **Physical dependence: Tolerance + withdrawal**

- **Intoxication:** after administration, effect of drug action on neuro-receptors occurs.
- **Withdrawal:** after ceased administration, an opposite effect of drug action occurs(if abusing CNS stimulant withdrawal is in form of sleepiness, if abusing CNS depressant withdrawal is in form of irritability or even fits) due to neuro-adaptation i.e. neurons decrease production of natural chemicals simulated by external drugs.
- **Tolerance:** after repeated administration, a given dose of drug produces a decreased effect due to neuro-adaptation i.e. down regulation of receptors

Types of psycho-active substances:

CNS stimulants - CNS depressants -Partial stimulants & partial depressants

a) CNS stimulants			
	Intoxication	**Withdrawal**	**Target receptors**
caffeine	Tremors, tachycardia	irritability	Adenosine
nicotine	Tremors, tachycardia	irritability	nicotine
mphetamines	Tremors, tachycardia	fatigue	dopamine
cocaine	Tremors, tachycardia	fatigue	dopamine
b) CNS depressants			
	Intoxication	**Withdrawal**	**Target receptors**
inhalants	Slurring, ataxia	Anxiety, fits	GABA
alcohol	Slurring, ataxia	Anxiety, fits	GABA
hypnotics	Slurring, ataxia	Anxiety, fits	GABA
opioids	Slow respiration	pains	Opioid
c) CNS partial stimulants - partial depressants			
	Intoxication	**Withdrawal**	**Target receptors**
cannabis	Apathy, delirium	flashbacks	cannabinoid
hallucinogens	Perceptual changes	flashbacks	glutamate

Etiology (nature and nurture):

Personal factors:

1. Genetic predisposition
2. Bipolarity, mood swings, personality disorders (borderline and antisocial are most common)
3. Poor problem solving and coping skills
4. Frustration intolerance

Social factors:

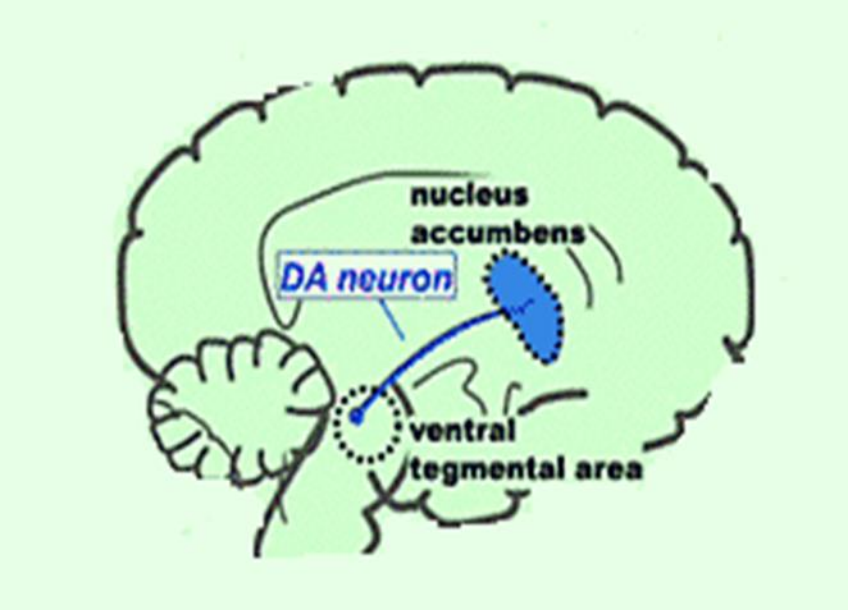
- Poor social support
- Drug availability
- social acceptability and modeling
- peer pressures

Biology of addiction

Repeated drug use produces changes in **brain-reward circuitry**(pleasure circuits). Psychoactive substances cause increase secretion of dopamine from ventral tegmental area in brain stem to toxic levels that scar nucleus accumbens so craving occurs with cues that provoke the emotional memory

Circuitry involved in addiction

(from Stahl Psychopharmacology)



Complications

Psychological complications:

Depression is related to CNS depressants, Psychosis is related to CNS stimulants

Dual diagnosis: this means the presence of both diagnoses of substance abuse and other psychiatric disorders.

Presence of psychiatric disorders increase risk of substance abuse, also substance abuse predispose to psychiatric disorders.

Integrated management plan to both improve the outcome and prognosis .

50% of schizophrenics, 60 % of manic, depressed abuse nicotine, cannabis, alcohol, 70% of anxiety patients abuse sedatives, alcohol.

Social complications:

Financial loses - Family troubles - Vocational troubles - Forensic troubles

Medical complications: Accidents, Infection, overdoses **Addiction medicine:**

1. Resistance to anesthesia and need for higher doses of anesthetics
2. Seizures, coma, neuropathy and myopathy
3. endocrinal disturbances
4. chest infection - endocarditis
5. dental caries - frequent bone fractures
6. skin infections (injections scars)

Treatment

Emergency care:

Overdose: Maintain vital signs - Use antidote –

Detoxification: Symptomatic measures to avoid withdrawal

Short term plan:
Remission induction:

- Motivational interviewing:

1. At 1st, patient is insightless about having a problem.
2. After being educated about his losses and being able to associate it to drug abuse, patient get frustrated saying "***I need change, but I can't change***"
3. After meeting X- addicts and being educated about remission rates "***I can change, but how?***"
4. Each patient ***tailor a recovery plan*** with help of mental health team suitable to his potentials and social support.
5. Plan is ***put into action*** with help of family or supporting key person.
6. Maintaining abstinence require educating patient about difference between ***lapse*** (abuse in spite of being motivated to stop) and ***relapse*** (totally giving up trying to stop).

Long term:
Relapse prevention:

1. disulfiram, naltrexone , or acamprosate, for alcoholism (**deterrence therapy**)
2. methadone, levomethadyl acetate (LAAM), or buprenorphine for heroin addiction
3. nicotine delivery devices or bupropion for tobacco dependence. (**replacement therapy)**
4. cocaine and nicotine **vaccines** under trial.

- **self-help groups** (Alcoholics Anonymous (AA) or other 12-step principles) and therapeutic community.

Impulse control disorder

- Pathological gambling -Intermittent explosive disorder Pyromania setting fires -Kleptomania: stealing: tension occur before one of these act, followed by guilt.
- persons with serotonin dys-regulation try to master anxiety
- Behavioral and psychodynamic psychotherapy +SSRIs

Forensic Psychiatry

- Crime targets mentally ill (easy victims) and lead to illness (PTSD). Mental illness decreases coping and increases violence.
- Psychiatrist in court: assess **mental capacity** (perceive-retain-weigh-believe-express data).
- Some plead ***not guilty for reason of insanity*** (not understand criminality of act) or ***guilty but mentally ill*** (irresistible impulse to commit crime- **mitigating factor-** even with a policeman at elbow of offender).
- Correctional psychiatry: prisoners have high rates of mental illness as substance abuse and antisocial personality. **Rehabilitation** decreases reoffending.

Vegetative disorder

Eating disorders

Obesity	**Body mass index=weight/height^2 Normally, it ranges between 20-25, 25-30in overweight, >30 in obesity. Complications: circulatory-skeletal**	**Endocrinal -Drugs- Bad eating habits: brain initiates eating due to distress not hunger**	**habit reversal training: eat only when hungry + stress coping diet - drugs -exercise – surgery**
Bulimia nervosa	Episodes of binge eating followed by compensatory behaviors: Fastening - exercise – enemas – laxatives – diuretics - vomiting (Purging type)	Unempathic family: critical and controlling - obsessive personality traits - depression	Family therapy-CBT-Psychodynamic therapy +SSRIs
Anorexia nervosa 7% die	Distorted body image (as fat), fear of weight gain, maintain body weight <85% of expected; by purging or restricting food complications: amenorrhea-osteoporosis- - bradycardia- lowT_3- cachexia-leukopenia-	Controlling family – obsessive traits - comorbid depression –	As Bulimia, but severe cases need Re-feeding in hospital to avoid metabolic complications

Sexual disorders

Unconsummated marriages: a couple who never had coitus (uninformed - inhibited) may reveal to gynecologist as infertility workup. Simple education has been markedly effective.

Sexual disorders may be secondary to medical or psychiatric disorders or primary:

1-Lack versus excessive of sexual **desire**

2-Dyspareunia / Vaginismus and Anorgasmia

3-Premature ejaculation and

4-Lack of genital response (Some men cannot be affectionate toward a woman except those whom they see as degraded (*Madonna Putana complex*).

- **Sex addiction: Don Juanism/Nymphomania)** Patient's life revolves around compulsive, repeated, increasing seeking out sex.

CBT - psychodynamic therapy + SSRIs can be helpful.

Egodystonic sexual orientation (homosexuality) is patient's distress due to preference of same sex as sexual partner. There is high comorbidity with gender identity disorder.

Gender identity disorder (trans-sexuality): dissatisfied with one's own sex. After 2 years of using opposite sex clothes, hormones patients who had reassignment surgery didn't improve with high rates of suicide

Disorders of sexual preference (Paraphilias) arousal by deviate stimuli or acts as: **pedophilia** (children), **gerontophilia** (old aged), **zoophilia** (animals), **necrophilia** (dead bodies) or sexual **sadomasochism** (cruel) **voyeurism** (peep on un-expecting naked people), **exhibitionism** (exposing oneself to un-expecting people), **scatalogia** (use phone/computers) **frotteurism** (rubbing oneself against un-expecting people), **oralism** (oral contact with genitals) **fetishism** aroused by objects that are intimate with the human body as shoes, gloves, and stockings), **transvestic fetishism** (wearing opposite sex clothes). Patients cope with the anxiety due to fear of sexual incompetence. Five interventions: external control (family / police) - reduction of sexual drives - treat comorbid (depression / anxiety) - CBT - psychodynamic therapy.

Sleep disorders

may be secondary to medical, psychiatric disorders or primary sleep disorders as:

Dysomnias:

Insomnia: difficult initiating (pain/anxiety) or maintaining sleep (depression /Obstructive sleep apnea or Restless leg syndrome)

Hypersomnia: excessive sleep. **Narcolepsy**: sleep attacks in the waking state

Sleep-wake schedule disorders like **jet lag disorder**.

Parasomnias:

Bruxism: teeth grinding during sleep. Bite plates protect teeth.

Enuresis: urination during sleep. Investigate infection, oxursis and spina bifida before fluid restriction, tricyclic antidepressants.

Sleep talking, walking and **terrors (**arousal early in night screaming, unresponsive to reassurance, then fall asleep again, forget this episodes) occur in nonREM (stages III, IV) **Nightmares** occur late in night during REM sleep and can be recalled, reassured.

Sleep Hygiene

☺ do

(1) Arise & go to bed **same** time & place daily

(2) comfortable sleep conditions

(3) evening **relaxation** routines (hot bath)

☹ don't

(1) daily **activity** in-bed (reading & studying)

(2) daytime **naps**

(3) evening **stimulating** activities, caffeine/nicotine, large meals, thinking of problems in bed

Thank you